

Unpacking Job Function Fit: A Multinomial Analysis of Personality, Demographics, and Career History

Zhuowei Deng, Romy Gou, Pinyi Li,
Elise Pham, Santsibyllah Seitz, Tingwei Wu

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Abstract

The goal of this study is to determine what traits and values align with different career types in order to determine an individual's ideal executive placement. This report explores the relationship between respondent psychometric scales and career function, as well as the predictive ability of demographic data for determining the job function of each respondent. The model can be used to detect mismatches, i.e. someone who values certainty and linearity may not thrive in a fast-paced ever-changing environment. The dataset used was provided by Dr. Dana Landis and contains a variety of demographic information such as age, gender, and country raised, professional experience information such as management level, career type, and time spent in current position, and respondent scores on two psychometric scales: one measuring common traits and one measuring the motivators of the respondents. After exploring the relationship between the different predictive variables with each other as well as their relationship with job function, two models were constructed as outlined above. It was determined that the traits which were the most important positive predictors for certain job functions were seeking challenge for Sales & Relationship, linear thinking for Engineering, charisma for Sales & Relationship, and agility for Software & Systems roles. The traits that most strongly decreased the likelihood of being in a role were linear thinking for Executive & General Management, agreeableness for Engineering, and agility for Sales & Relationship. The demographic information that was most significant in terms of distinguishing whether a respondent was an engineer or one of the other top ten job functions was that related to gender and education level. Ultimately, we concluded that there does seem to be a relationship between the top ten job functions with the personality traits, motivators, and demographic data of the respondents.-

1 Introduction

1.1 Research Questions and Goals

In today's workforce, understanding the factors that influence a person's career path is more important than ever. This project investigates the relationship between psychometric traits, demographic background, and job function. Our primary research questions are:

1. Which psychometric traits are most predictive of specific job functions?
2. How do demographic and professional background factors (e.g., gender, education, country raised/employed) contribute to job function classification?

The goal of this analysis is to build classification models that can help interpret how individual characteristics and background relate to professional roles. By identifying the most influential predictors for each job function, insight can be provided into the psychological and background profiles commonly associated with different career paths.

1.2 Background

Career decisions are shaped by a combination of internal traits and external circumstances. Many occupational psychology frameworks reinforce the idea that personality traits

and environmental factors jointly determine job fit and performance. One influential lens is Trait Activation Theory, which holds that individuals display certain traits only when job situations “activate” them — for instance, decisiveness manifests in high-stakes decision roles, whereas social traits emerge in team-driven environments [3]. This theory highlights why specific roles demand distinct trait profiles, making psychometric measures useful for predicting job success.

Closely related is the Person–Job Fit framework, which asserts that alignment between an individual’s characteristics and job requirements leads to better engagement, innovation, and retention [1]. Empirical studies find that employees whose personal values and traits match their roles are more involved and innovative, mediated by job engagement and moderated by career commitment.

In this context, our study analyzes a dataset containing self-reported psychometric traits, job function labels, and demographic indicators, focusing on the 10 most common job functions. Applying both Trait Activation and Person–Job Fit theories, we investigate whether specific traits, such as agility, linearity, or charisma, are predictive of particular roles, and how demographic and professional background factors contribute to job classification. This dual-framework approach aims to uncover how personality and context interact to influence career paths and support more effective role alignment in organizations.

2 Data Description

2.1 Dataset Properties

The data used to complete this project was provided by Dr. Dana Landis and included information relating to respondent demographics, employment information, personality trait scores, and motivators. There are 229770 observations and 23 variables.

- Variable of Interest: **function**.
 - The category of career of the respondent
 - Only the top ten job functions with the most respondents were used: Engineering, Executive & General Management, Finance/Accounting, HR & Talent, Marketing & Brand, Operations, Procurement & Logistics, Program/Project Leader, Sales & Relationship, and Software & Systems
- Personality Trait Variables:
 - **scaleA_Agile**: A measure of respondent’s agility ranging from approximately -4 to 4.
 - **scaleA_Charisma**: A measure of respondent’s charisma ranging from approximately -4.5 to 3.
 - **scaleA_Aspiring**: A measure of respondent’s level of aspiration ranging from approximately -4 to 3.
 - **scaleA_Agreeable**: A measure of respondent’s agreeableness ranging from approximately -4 to 3.5.

- **scaleA_Optimism**: A measure of respondent’s optimism level ranging from approximately -5 to 3.
- Driver Variables:
 - **scaleD_WLB**: A measure of how much a respondent seeks work-life balance ranging from about -3 to 5.
 - **scaleD_CHA**: A measure of how much a respondent seeks challenge ranging from about -4 to 3.
 - **scaleD_COO**: A measure of how much a respondent seeks cooperation ranging from about -4.5 to 3.5.
 - **scaleD_AUT**: A measure of how much a respondent seeks autonomy ranging from about -4.5 to 3.5.
 - **scaleD_POW**: A measure of how much a respondent seeks power ranging from about -4 to 4.
 - **scaleD_LIN**: A measure of how much a respondent seeks linear processes ranging from about -3 to 4.

Note: Psychometric variables such as **scaleD_AUT** and **scaleA_Agreeable** were produced using a forced choice scale and have been normalized to Z-scores:

$$z = \frac{x - \mu}{\sigma}$$

While all respondents filled out the Scale A personality trait assessment, about 16% of the respondents did not complete the Scale D (motivators) assessment.

The first model used the personality traits and driver variables to predict job function. The second model used **countryEmployed**, **countryRaised**, **education** (4-Year, Masters, Doctorate, Some College, Associates, High School, No High School, or Some High School), **gender** (only Male and Female considered), **managementLevel** (C-level, Executive, High-level, Low-level, Mid-level, or None), **yrsManaging**, **yrsCurrentPosition**, and **age** (values outside of 18-70 filtered out) to predict **function**.

2.2 Exploratory Data Analysis

Before creating the models, exploratory data analysis was performed in order to determine the distribution of the variables, any possible relationship across predictor variables, and the extent of data cleaning necessary before constructing the models.

We directly plotted the distributions of the psychometric scale variables to examine their shapes, and as shown in Figure 1, most trait scores exhibit approximately bell-shaped distributions. Each trait, such as **scaleA_Agile**, **scaleA_Agreeable**, **scaleD_POW**, and others, was centered and scaled, resulting in distributions with means near zero and standard deviations of approximately one.

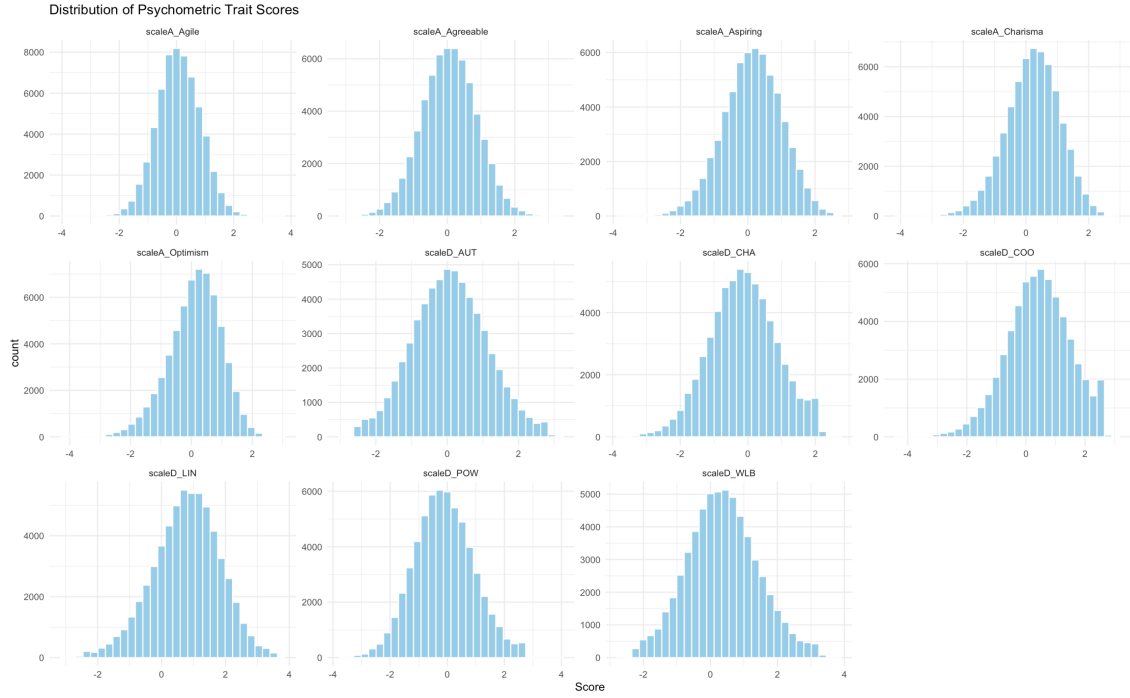


Figure 1: Plot visualizing the distribution of psychometric trait scores

We can also observe in Figures 2 and 3 that the distributions of psychometric trait scores—including both personality (A-Traits) and motivational drivers (D-Traits)—are relatively consistent across the top ten job functions. Although there are slight shifts in medians and variability for some roles, the general shape and spread of the boxplots suggest that no single job function is overwhelmingly defined by a unique trait profile.

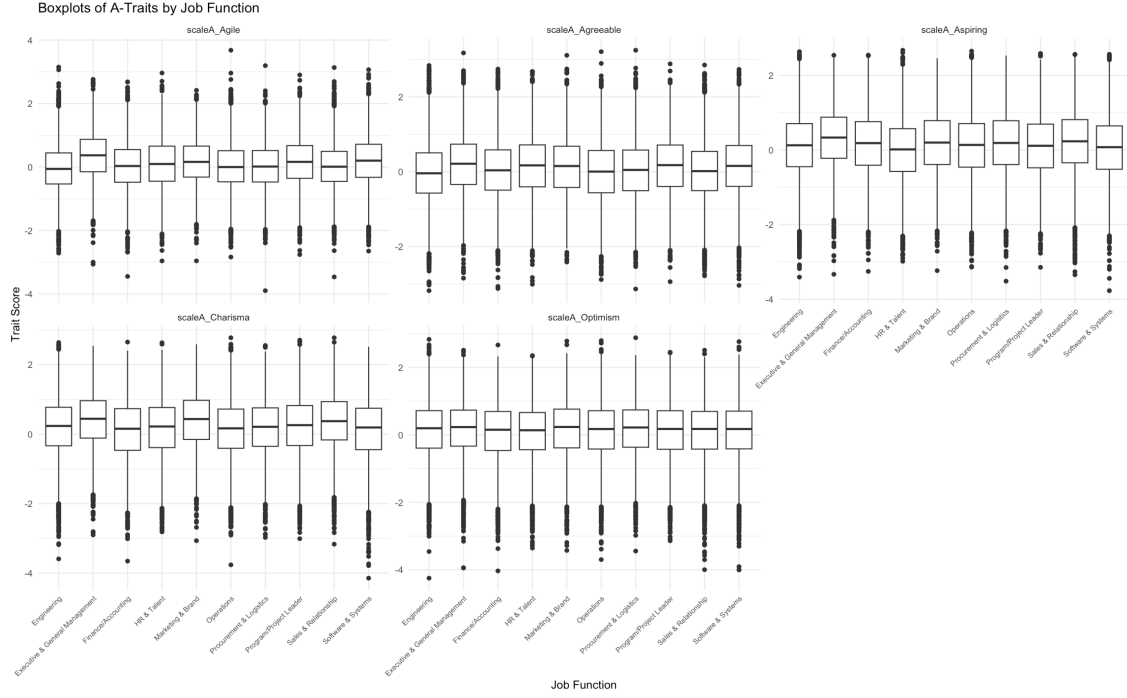


Figure 2: Plot visualizing the distribution of personality trait scores by top ten job function

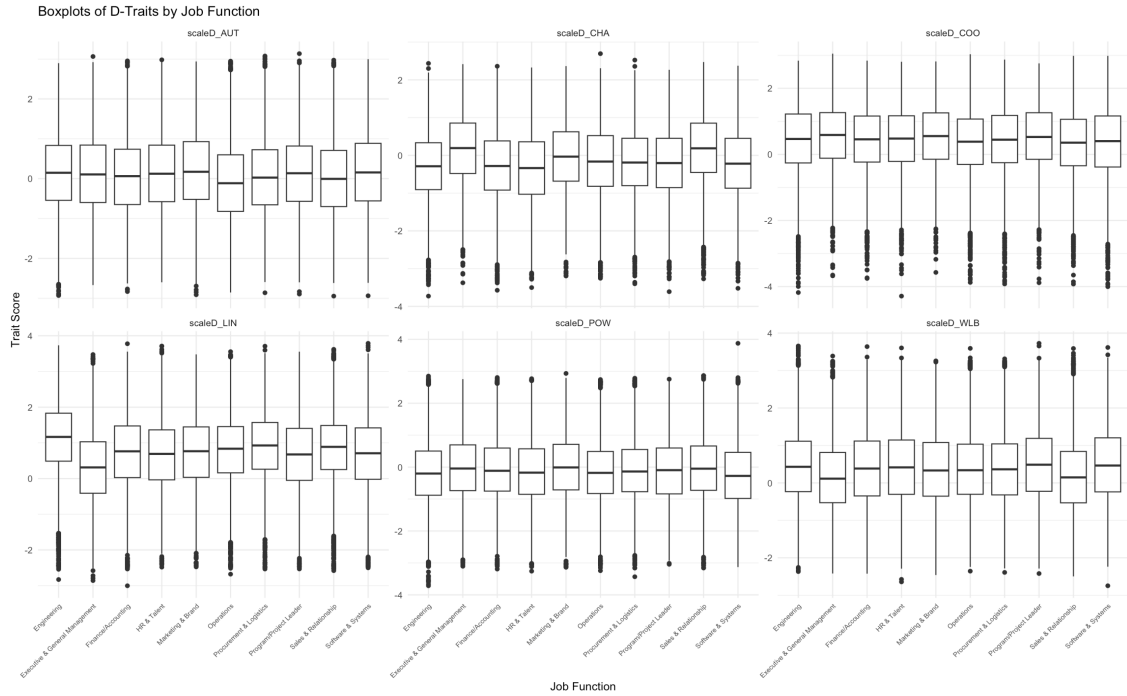


Figure 3: Plot visualizing the distribution of motivator and drivers scores by top ten job function

Additionally, as shown in Figure 4, none of the psychometric traits exhibit strong pairwise correlations. This lack of multicollinearity is important, as it supports the validity of

including all traits as independent predictors within the same classification model.

Correlation Heatmap of Psychometric Traits

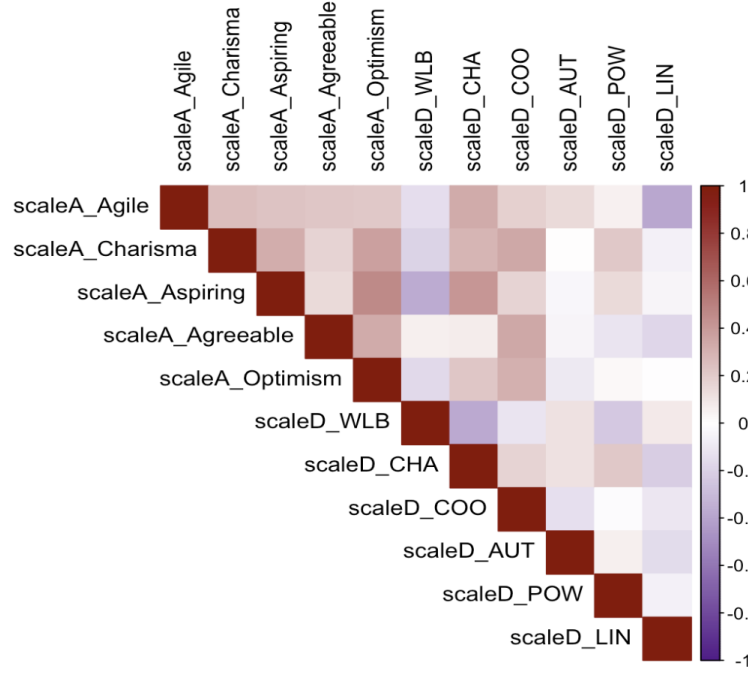


Figure 4: Plot visualizing correlation between Scale A personality traits and Scale D drivers

For our second model, referred to as the non-traits model, we examined how demographic and professional background variables might predict job function. As an initial exploratory step, we identified the top three most common job functions by country of origin among respondents from the ten countries with the highest response counts. These results are displayed in Figures 5 and 6. This analysis highlights how job function distributions can vary across cultural and national contexts, suggesting potential links between occupational trends.

Top 3 Job Functions in Countries Ranked 1 to 5 by Respondent Count					
Rank	United States of America	Turkey	Brazil	India	United Kingdom
1	Operations (12.5%)	Engineering (28.4%)	Sales & Relationship (16.4%)	Engineering (18.6%)	Sales & Relationship (10.6%)
2	Sales & Relationship (11.2%)	Finance/Accounting (7.8%)	Engineering (9.3%)	Operations (12.1%)	Executive & General Management (9.0%)
3	Software & Systems (8.0%)	Software & Systems (7.0%)	Operations (7.8%)	Software & Systems (8.9%)	HR & Talent (8.9%)

Figure 5: Table showing the percentage of respondents with the top three most common job functions by country raised, for countries ranked 1 to 5 by total number of responses

Top 3 Job Functions in Countries Ranked 6 to 10 by Respondent Count					
Rank	Australia	Mexico	China	Italy	Sweden
1	Sales & Relationship (10.5%)	Sales & Relationship (12.7%)	Sales & Relationship (13.0%)	Sales & Relationship (9.7%)	Engineering (15.8%)
2	Claims Management (7.8%)	Operations (10.5%)	Engineering (9.4%)	Operations (8.7%)	Operations (10.7%)
3	Executive & General Management (7.8%)	Engineering (7.8%)	Medical Research (7.6%)	Program/Project Leader (8.3%)	Medical Research (9.6%)

Figure 6: Table showing the percentage of respondents with the top three most common job functions by country raised, for countries ranked 6 to 10 by total number of responses

Following this, we explored the distributions of the three key numeric predictors in our dataset: age, years in current position, and years of management experience. The age appeared to be approximately distributed normally, while both the current years and the years of management showed a noticeable skew to the right 7.

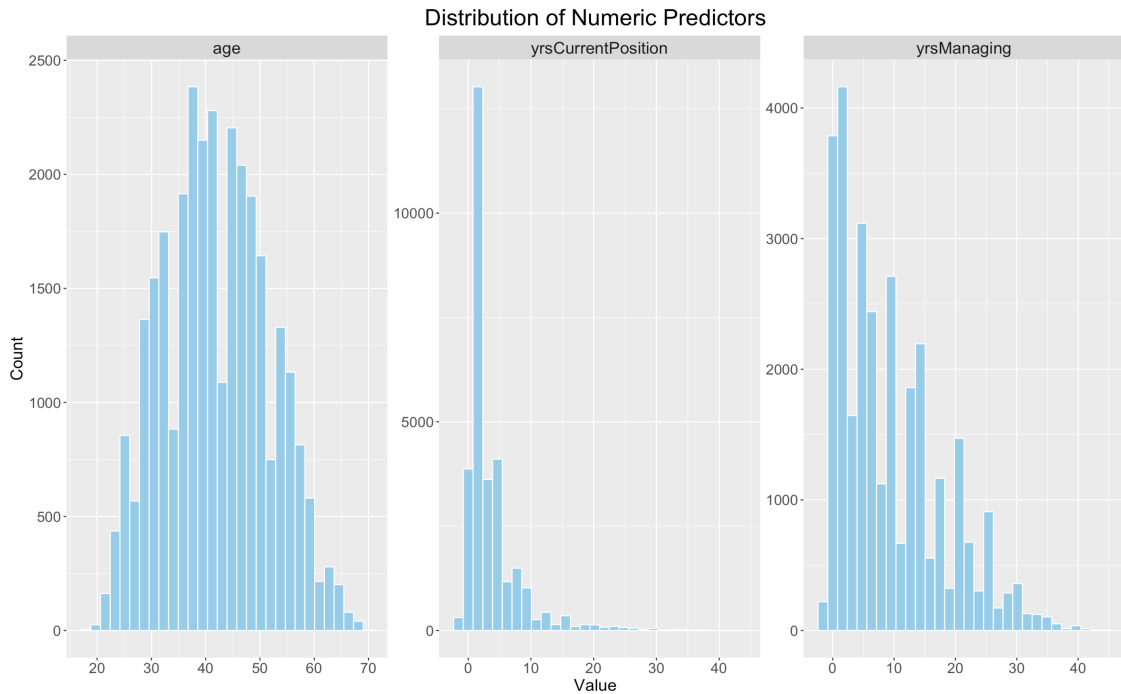


Figure 7: Histograms showing the distributions of numeric predictor variables

Additionally, we observe that age, years in current position, and years managing are all positively correlated with one another as expected given their natural interdependence. Age and years managing showed a moderate positive correlation of 0.67, while age and years in current position had a weaker correlation of 0.28. Years managing and years in current position were only weakly correlated, with a coefficient of 0.20.

To further explore these relationships, we created boxplots in Figures 8, 9, and 10 to examine how these numeric variables are distributed across the top ten job functions. These plots reveal that the distribution of age, tenure in current position, and years of management

experience varies noticeably by job function. For example, executive and managerial roles tend to be associated with older respondents and longer tenure, while technical roles like Engineering and Software & Systems show a younger and more diverse distribution.

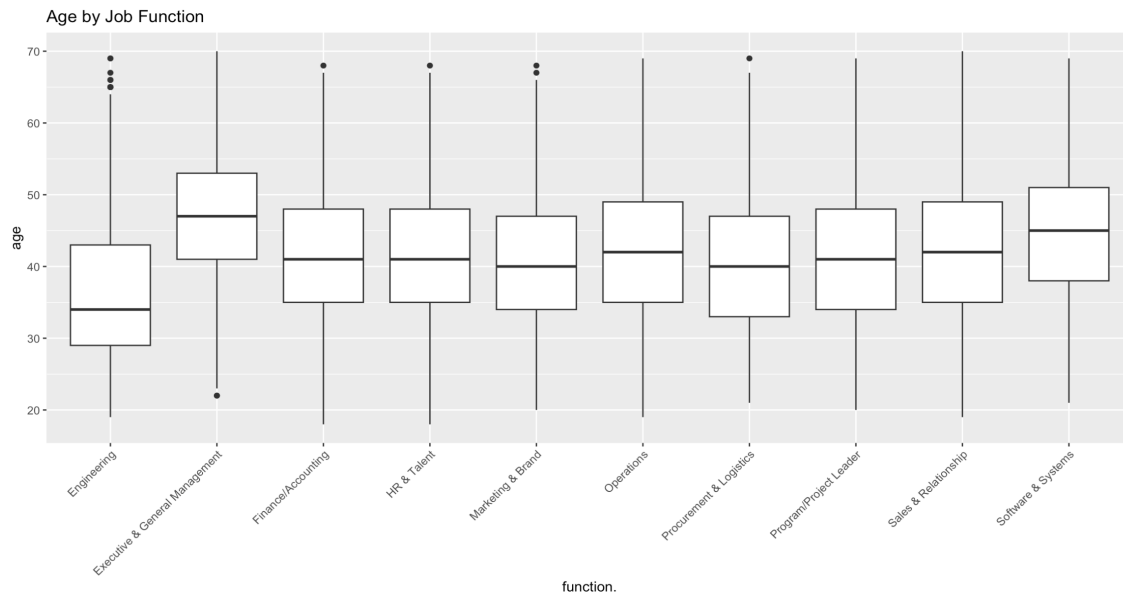


Figure 8: Boxplots showing the age distribution by top ten job function

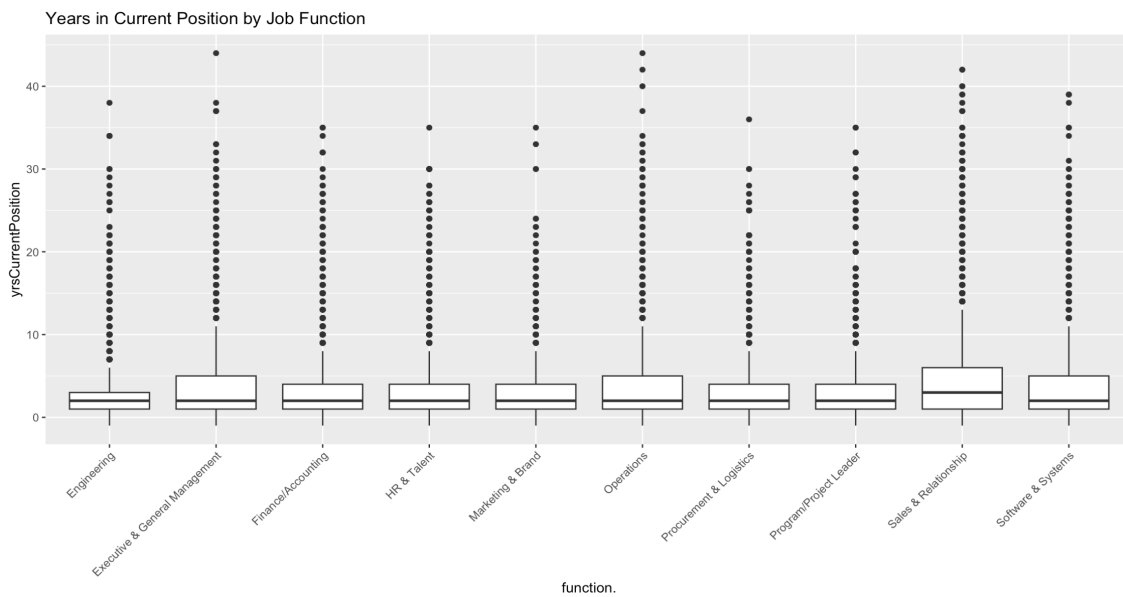


Figure 9: Boxplots showing the distribution of years spent in current position by top ten job function

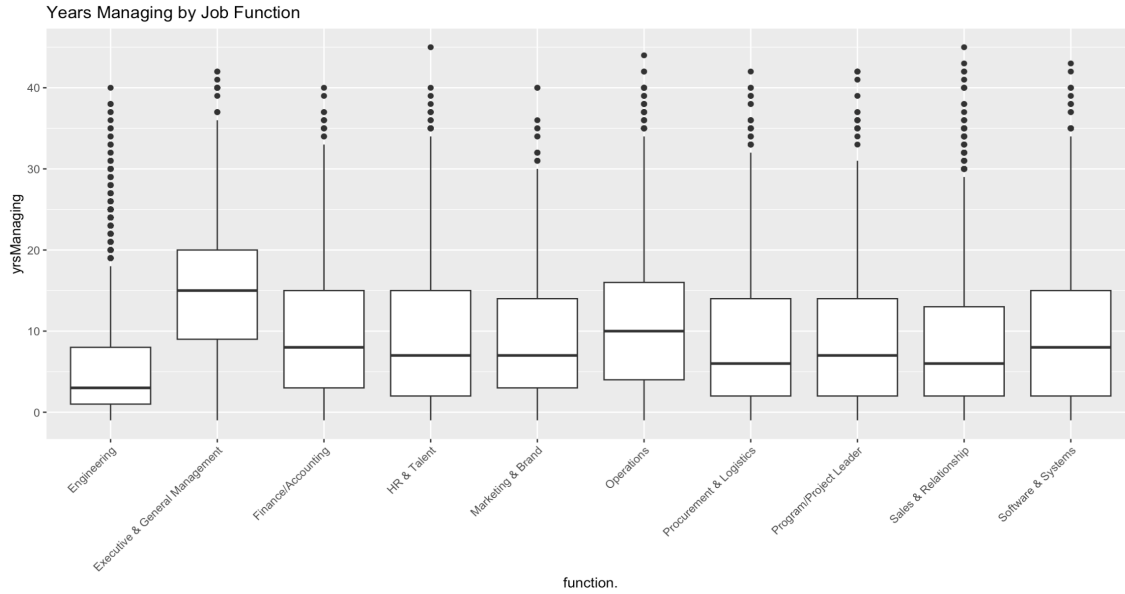


Figure 10: Boxplots showing the distribution of years spent managing by top ten job function

Additionally, as shown in Figures 11, the bar charts depicting the distribution of **gender**, **education**, and **managementLevel** across the top ten job functions reveal noticeable variation in proportions. For example, individuals in C-level and executive roles are disproportionately represented in Executive & General Management and Engineering job functions, suggesting that these functions are more prevalent among senior professionals. In terms of gender, we observe that Software & Systems and Engineering roles have higher representation among male respondents, whereas HR & Talent and Marketing & Brand show a relatively more balanced or even female-leaning distribution.

With respect to education, job functions such as Software & Systems and sales & Relationship appear consistently across most educational levels, while functions like Engineering and Finance/Accounting show stronger representation among respondents with advanced degrees such as Doctorates or Master's. These patterns suggest that demographic and educational characteristics may be predictive of job function.



Figure 11: Bar charts showing the proportion of each education level, management level, and gender taken up by each of the top 10 job functions.

3 Methods

We aim to use both multinomial logistic regression with Lasso regularization and standard multinomial logistic regression to identify the predictors most strongly associated with job function classification.

3.1 Data Preprocessing

The data preprocessing steps involved several stages to prepare the dataset for analysis. First, an age column was added by calculating the difference between the response date and the year of birth. The dataset was then filtered to include only respondents aged between 18 and 70. Any missing age values were imputed with the mean age of 38. Next, the dataset was restricted to respondents from the top 10 raised countries by number of responses. The data was then split into two subsets: one containing psychometric traits (Scale A and Scale D) and the other containing non-trait variables (demographic and career-related features). Redundant columns such as row number, respondent ID, year of birth, and date were removed. In the non-trait dataset, outliers were excluded by filtering out records where the years of management experience or years in the current position were less than zero. Finally, records containing missing or invalid values, including NA (Not Available), NR (No Response), None, or empty strings (""), were removed to ensure data quality.

3.2 Model Fitting

Psychometric Traits

A histogram was plotted for each psychometric trait to examine the distribution of scores. While most traits exhibited an approximately normal distribution, slight deviations at the tails were observed—indicating mild skewness or heavier tails, which are common in real-world data. As these deviations were not severe, no further data transformations were applied.

To assess relationships between traits, a Pearson correlation matrix was computed and visualized using a heatmap. The analysis revealed weak or moderate correlations within subsets of traits, but no evidence of strong multicollinearity. As a result, no traits were removed due to redundancy.

We modeled the relationship between psychometric traits and the top 10 job functions using multinomial logistic regression with Lasso (L1) regularization. The Lasso penalty encourages sparsity in the coefficient estimates, aiding feature selection across multiple outcome classes. The optimal regularization parameter λ was determined via cross-validation using `cv.glmnet`. The top three predictors with the largest coefficients were selected per job function to highlight the strongest associations identified by Lasso regularization.

Non-Trait Variables

A standard multinomial regression model was fitted to assess the influence of demographic and career-related factors on the top 10 job functions within the top 10 countries of employment. The predictor matrix X included factors such as gender, management level, education, years in position, and country of employment and upbringing. The response variable y was the factor-encoded job function. This model estimated the log-odds of each job function relative to the baseline category (Engineering), with all predictors included simultaneously. For each categorical predictor, one level was omitted and used as the reference category, with all coefficients interpreted relative to that baseline. Coefficient estimates and their standard errors were used to compute z-scores and p-values. The p-values were then used to identify the top 3 non-trait variables that were statistically significantly in predicting different job functions.

Predictor	Baseline (Reference Level)
Job Function (Y)	Engineering
gender	F
education	4-Year
managementLevel	C-level
countryEmployed	Australia
countryRaised	Australia

Table 1: Reference levels for categorical variables in the multinomial logistic regression model.

Generally, we use LASSO to select variables for better prediction and handle high-dimensional data by shrinking irrelevant coefficients to zero; we use `multinom()` to perform multinomial logistic regression for inference, testing the statistical significance of predictors on categorical outcomes.

4 Results

Figure 12 displays a heatmap of trait coefficients for each job function based on a multinomial LASSO regression. Warmer colors (toward red) indicate positive coefficients—traits that increase the likelihood of a respondent belonging to a specific job function—while cooler colors (toward purple) reflect negative associations.

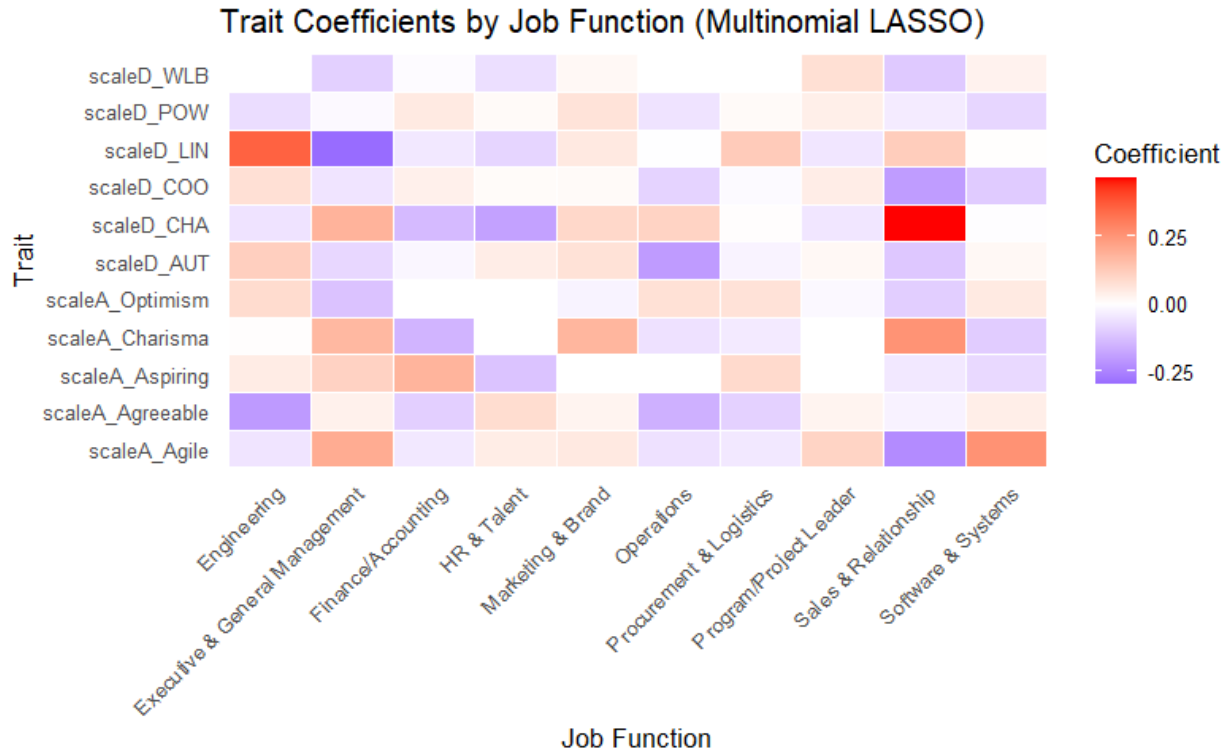


Figure 12: Heatmap showing the trait coefficients from a multinomial LASSO model across the top ten job functions

Overall, it is observed that certain psychometric traits are more strongly associated with specific types of careers. For instance, individuals who score high in **Charisma** and **Challenge-seeking** are more likely to work in **Sales & Relationship Management**, a role that typically requires persuasive communication and frequent, high-stakes client interaction. Similarly, **Agility**, representing adaptability and comfort with uncertainty, is notably prominent among individuals in **Software & Systems** roles. This finding likely reflects the need for flexibility in the face of rapidly evolving technologies and problem-solving in unpredictable environments. We observe positive associations for traits such as seeking **challenge**, **charisma**, and **agility** in **Executive & General Management** roles. This pattern aligns with the expectations of leadership positions, where dynamic thinking, strong interpersonal influence, and adaptability are critical for guiding teams and driving organizational strategy.

In addition, those in **Engineering** are more likely to exhibit **linear thinking**, reflecting the need for analytical, step-by-step problem solving. Interestingly, respondents in **Executive and General Management** roles tend to show a negative association with linear

thinking. These roles require a broader perspective, focusing on long-term strategy and organizational direction rather than sequential tasks.

Some of the negative associations between traits and job functions highlight possible mismatches between personality and professional demands. For instance, individuals with high levels of **agreeableness** are less likely to be in **Engineering**, possibly because engineering roles emphasize objective analysis and technical precision over interpersonal sensitivity. Likewise, lower representation of **agility** in **Sales & Relationship** roles may indicate that excessive flexibility could hinder success in positions that depend on stability, consistency, and the cultivation of long-term client relationships. We also observe that lower levels of **seeking challenge** are linked to **HR/Talent** roles—perhaps reflecting that these positions prioritize empathy, routine processes, and organizational stability over risk-taking. In the **Operations** function, negative associations with **agreeableness** and **autonomy** may reflect the need for structured coordination and compliance with established procedures rather than independent or overly accommodating tendencies.

To better understand the relationship between psychometric traits and career specialization, we examined the top three positive trait coefficients from our LASSO model for each job function. These predictors reflect the traits most strongly associated with an increased likelihood of belonging to a given job category. As shown in the table 13 above, specific traits consistently emerged for distinct roles.

Job Function	Rank1	Rank2	Rank3
Engineering	scaleD_LIN	scaleD_AUT	scaleA_Optimism
Executive & General Management	scaleA_Agile	scaleD_CHA	scaleA_Charisma
Finance/Accounting	scaleA_Aspiring	scaleD_POW	scaleD_COO
HR & Talent	scaleA_Agreeable	scaleA_Agile	scaleD_AUT
Marketing & Brand	scaleA_Charisma	scaleD_CHA	scaleD_AUT
Operations	scaleD_CHA	scaleA_Optimism	scaleD_WLB
Procurement & Logistics	scaleD_LIN	scaleA_Aspiring	scaleA_Optimism
Program/Project Leader	scaleA_Agile	scaleD_WLB	scaleD_COO
Sales & Relationship	scaleD_CHA	scaleA_Charisma	scaleD_LIN
Software & Systems	scaleA_Agile	scaleA_Optimism	scaleA_Agreeable

Figure 13: Table showing the psychometric traits coefficients from a multinomial LASSO model across the top ten job functions

To complement our analysis of psychometric traits, we also explored how demographic and professional background factors contribute to job function classification. Figure 14 displays a heatmap of coefficients derived from a multinomial logistic regression model using

non-trait predictors such as education level, gender, management level, country raised/employed, age, and experience.

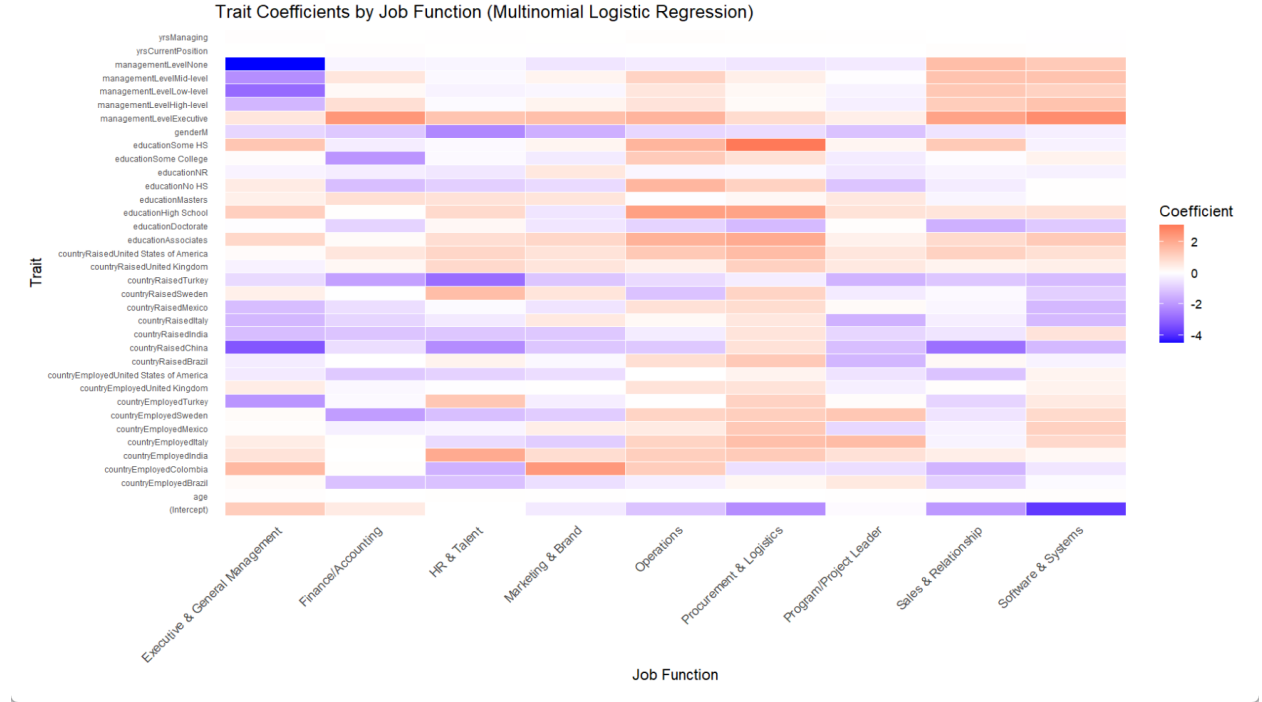


Figure 14: Top three psychometric predictors ranked by their importance in distinguishing each job function based on the multinomial logistic regression model

When using job function as the outcome variable and setting Engineering as the baseline, our multinomial logistic regression highlights the demographic predictors that most distinguish each function from engineering. In other words, these top variables distinguish each listed job function from Engineering, highlighting demographic information that makes individuals more likely to belong to one role over another. For example, Executive & General Management roles are significantly associated with being male and occupying low- to mid-level management positions. In Finance & Accounting, higher educational attainment—specifically having a master’s degree—stands out alongside male gender and executive-level management. For HR & Talent, being male, holding a master’s degree, and being older increased the likelihood of belonging to that function. Marketing & Brand similarly showed strong ties to having a master’s degree, male gender, and an executive-level title. In Operations, top predictors included holding an associate’s degree or high school diploma, while male gender remained relevant. These patterns reflect broader findings in the literature that emphasize how organizational structures and gender biases shape role assignment and career progression across industries [2].

Job Function	Rank 1	Rank 2	Rank 3
Executive & General Management	Gender M	Management Level Low-level	Management Level Mid-level
Finance/Accounting	education Masters	Gender M	Management Level Executive
HR & Talent	Education Masters	Gender M	age
Marketing & Brand	Education Masters	Gender M	Management Level Executive
Operations	Education Associates	Education High School	Gender M
Procurement & Logistics	Education Associates	Education High School	Education Some HS
Program/Project Leader	Education Masters	Gender M	Age
Sales & Relationship	Education Doctorate	Gender M	Age
Software & Systems	(Intercept)	age	Education Associates

Figure 15: Table showing Top 3 Predictors Differentiating Each Job Function from Engineering

Overall, gender and education level were the most significant demographic factors in differentiating engineers from individuals in the other top ten job functions. One or the other appeared in every job function as a key differentiator. Surprisingly, country of employment and country raised were not significant in the classification model, suggesting that geographic background may play a less prominent role in determining career paths compared to personal characteristics and demographic traits.

5 Conclusions, Limitations, and Advice

5.1 Limitations

- **Missing Data:** About 16% of the responses for Motivation (Scale D) are not available, and a review of the data reveals that 50% of management experience (yrsManaging) data is incomplete. This sample size reduction may be a contributing factor to potential bias, as individuals who opt to respond to a particular question may not be representative of the overall population.
- **Sample Selection:** The analysis focuses exclusively on the ten job functions. It has been observed that countries may neglect to acknowledge the significance of relation-

ships that deviate from conventional roles. The concept of "regions" is also applicable in this context.

- **Reference Group Effect:** The reference group effect was incorporated into the multinomial logistic regression model by setting Engineering as the reference group. The reference category is employed herein. Consequently, all coefficients and interpretations are relative. It is important to consider the implications of this on the field of engineering, as this may result in a limitation of our understanding of the full spectrum of differences among the various job functions.
- **Model and Interpretation:** Multinomial LASSO regression was utilized for personality analysis. The selection of the most significant variables is facilitated by the identification of motivation traits. However, LASSO does not provide traditional significance tests (e.g., p-values), which precludes the possibility of conducting traditional significance tests. There is a definitive conviction regarding the significance of each trait. A model was developed to analyze the impact of demographic factors. Conversely, this approach precludes the assessment of the interaction between personality and demographics.
- **Cross-Sectional Design:** Given the nature of our data, which is a "snapshot" in time, it is not possible to ascertain if the relationship between traits and job selection, or the role of job selection in shaping traits over time, is a subject of interest in the field of sociology.
- **Measurement Issues:** The subject of full documentation on the personality was not fully developed. Motivation scales were subjected to calculation and standardization. This potential alteration could have ramifications for the reliability of the system.

5.2 Conclusions

The study demonstrates that respondents' personality traits, motivational factors, and job function are systematically correlated with demographic characteristics and job function. To elaborate further:

- **Trait Effects:** It has been demonstrated that challenge-seeking (scaleD_CHA) and charisma (scaleA_Charisma) are robust and positively correlated with roles involving sales and relationship management. Linear thinking (scaleD_LIN) is most evident among engineering professionals, as evidenced by the significant correlation between this variable and profession. There is a strong correlation between these individuals and those in executive positions. Agility (scaleA_Agile) is a particularly pertinent consideration. The correlation between software and systems roles and sales functions is demonstrated to be inverse.
- **Demographic Effects:** The most significant factors identified were gender and education level. The impact of country or culture is negligible in comparison to that of demographic predictors of job function. The location at which the work is performed

is hereby designated as "current work location." For instance, holding a master's degree or being male has been demonstrated to be associated with a higher probability of securing employment in roles related to human resources, talent management, or operations.

The findings indicate that psychometric assessments provide significant value in the context of employment. The process of matching and talent placement is of paramount importance. It is recommended that organizations consider incorporating personality and motivation testing into the recruitment and leadership development processes. The objective is twofold: first, to enhance the fit, and second, to mitigate the risk of a job-role mismatch. Nevertheless, further research is necessary to validate the causal relationships between these traits and career outcomes. It is imperative to explore a wider variety of functions and cultural backgrounds.

5.3 Advice for Future Work

Future work will benefit from a more exhaustive data set which covers additional specific job functions and countries, so that more extensive and reliable results can be produced. It is imperative to consider the integration of personality traits and demographics in the modeling process to facilitate the exploration of potential outcomes. The following section will address the concept of interactions. The utilization of longitudinal surveys is imperative in order to ascertain the extent to which jobs and traits exert a reciprocal influence over the course of time.

In summary: The study demonstrates unequivocal associations between particular personality and motivational traits. The analysis encompasses both job functions and the value of demographic factors, including education and gender. Despite the presence of certain constraints, these findings can provide a foundation for support for employers and professionals. The objective is to facilitate the development of optimal career trajectories and hiring practices.

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